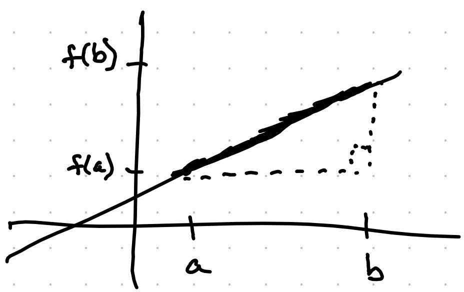
The arc length formula – 2.4

Goal: find the length of a curve.

We can do this when dealing with a line!

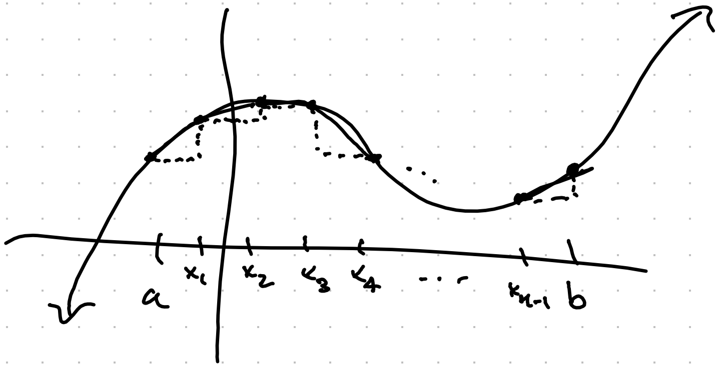


Pythagorean theorem:

$$\text{length} = \sqrt{(b-a)^2 + (f(b)-f(a))^2}$$

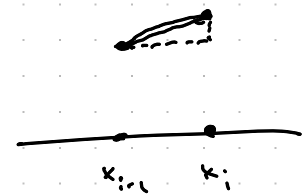
If we have something curved instead:

try to zoom in and approximate!



Divide $[a,b]$ into n pieces

Zooming in on $[x_{i-1}, x_i]$ looks like



Length of the curve from $(x_{i-1}, f(x_{i-1}))$ to $(x_i, f(x_i))$ is approximately equal to the length of the hypotenuse.

$$\left((x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2 \right)^{1/2}$$

Each interval contributes $\left((x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2\right)^{1/2}$

$$= \underbrace{(x_i - x_{i-1})}_{\Delta x}\left(1 + \left(\frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}}\right)^2\right)^{1/2}$$

By the Mean Value Theorem, there is some $x_i^* \in [x_{i-1}, x_i]$ such that $\dfrac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} = f'(x_i^*)$

Length contribution from $[x_{i-1}, x_i]$ is $\sqrt{1 + f'(x_i^*)^2}\,\Delta x$

Total length $\approx \displaystyle\sum_{i=1}^{n} \sqrt{1 + f'(x_i^*)^2}\,\Delta x$, which is

a Riemann sum!

Length is approximately $\sum_{i=1}^{n} \sqrt{1+f'(x_i^*)^2}\,\Delta x$,

and this should approach the actual length

as $n \to \infty$. Thus, define the length to be

$$\boxed{\text{Length} = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{1+f'(x_i^*)^2}\,\Delta x = \int_a^b \sqrt{1+f'(x)^2}\,dx}$$

"arc length formula"